

Chapter 4, Problem 84P

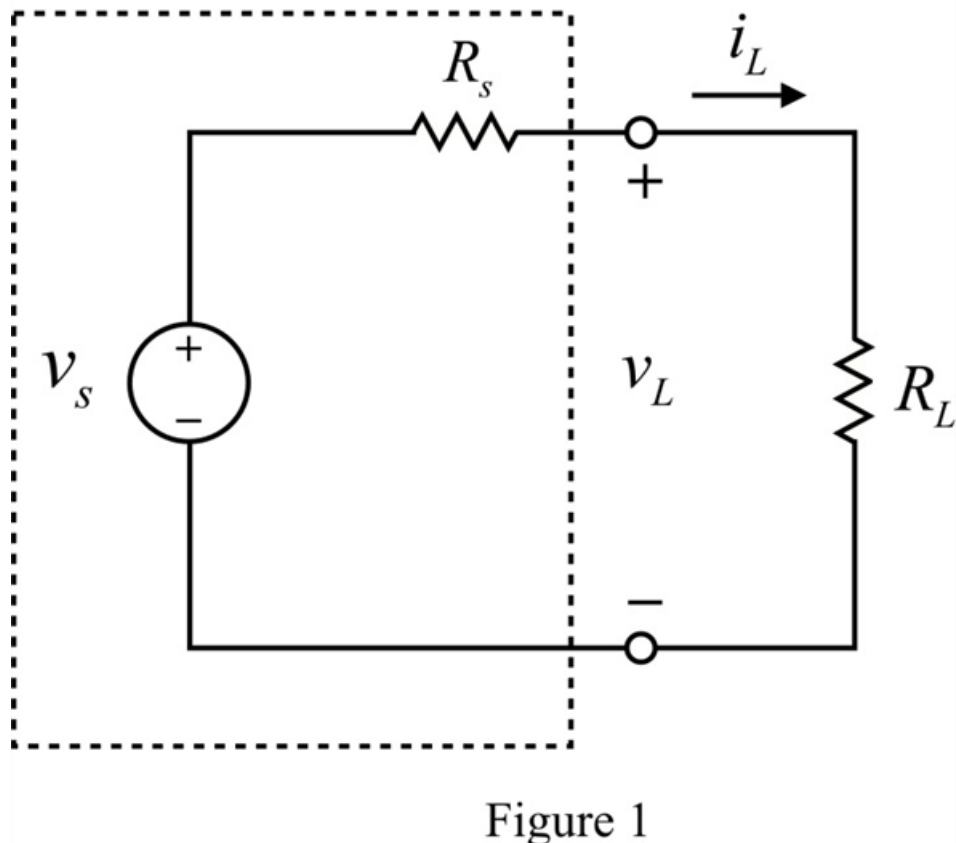
Problem

When connected to a $4\text{-}\Omega$ resistor, a battery has a terminal voltage of 10.8 V but produces 12 V on an open circuit. Determine the Thevenin equivalent circuit for the battery.

Step-by-step solution

Step 1 of 4

Draw the following circuit diagram.



Step 2 of 4

Consider the open circuit voltage.

$$v_s = v_{oc} = 12\text{ V}$$

Therefore, the Thevenin voltage is 12 V .

Step 3 of 4

The terminal voltage when load R_L is connected is,

$$v_L = 10.8 \text{ V}$$

Apply Ohm's law to the load of the circuit shown in Figure 1.

$$v_L = i_L R_L$$

$$i_L = \frac{v_L}{R_L}$$

Substitute 10.8 for v_L , and 4 for R_L .

$$\begin{aligned} i_L &= \frac{10.8}{4} \\ &= 2.7 \text{ A} \end{aligned}$$

Step 4 of 4

Apply Kirchhoff's voltage law to the circuit shown in Figure 1.

$$v_s - i_L R_s - i_L R_L = 0$$

$$R_s = \frac{v_s - i_L R_L}{i_L}$$

Substitute 12 for v_s , 2.7 for i_L , and 4 for R_L .

$$\begin{aligned} R_s &= \frac{12 - 2.7 \times 4}{2.7} \\ &= \frac{1.2}{2.7} \\ &= 0.444 \Omega \end{aligned}$$

Therefore, the Thevenin resistance is 0.444Ω .

Draw the Thevenin equivalent circuit for the battery as shown in Figure 2.

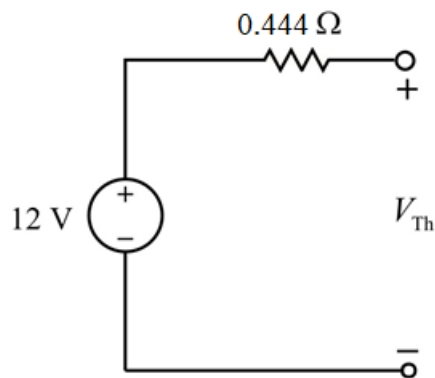


Figure 2